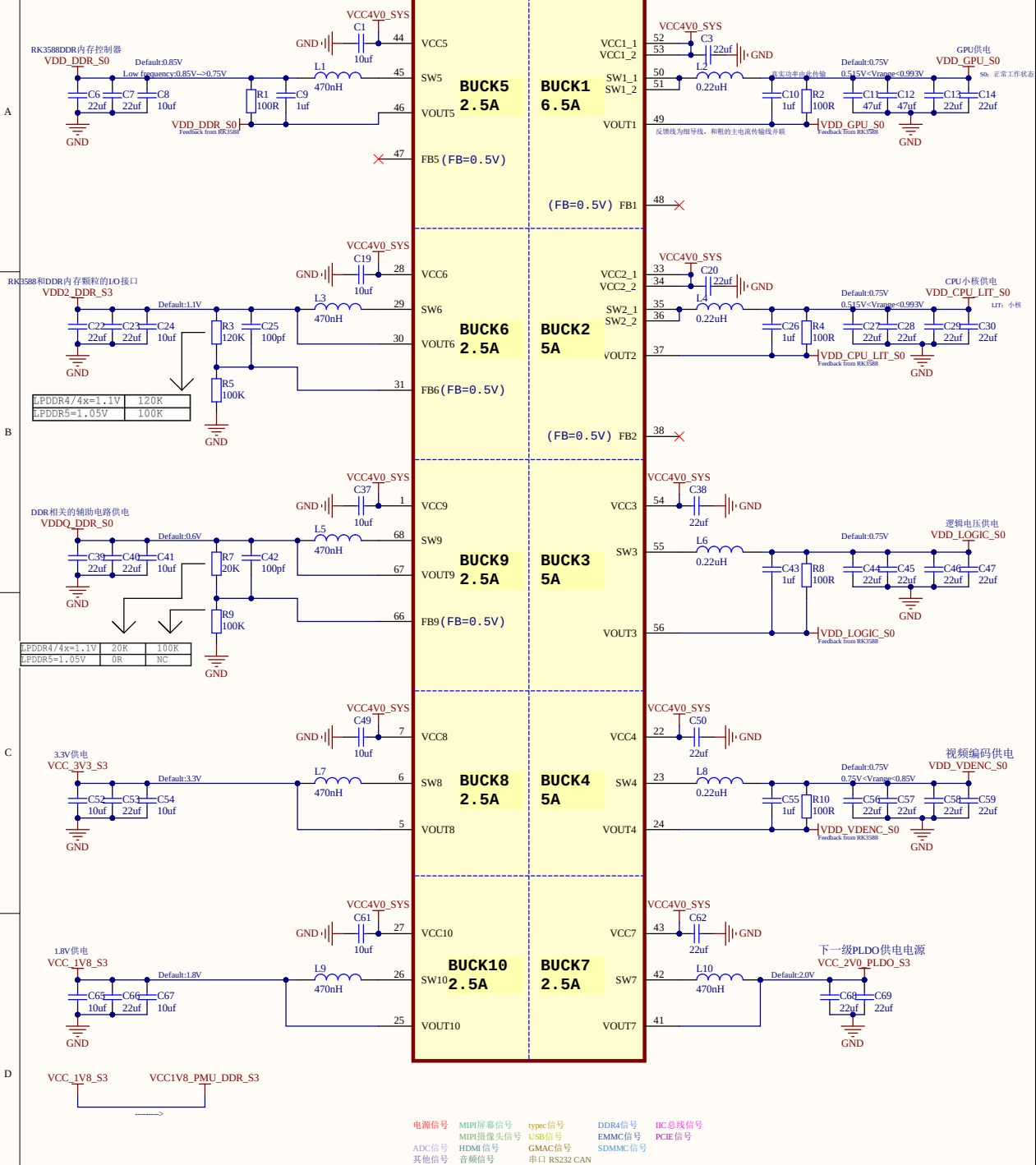
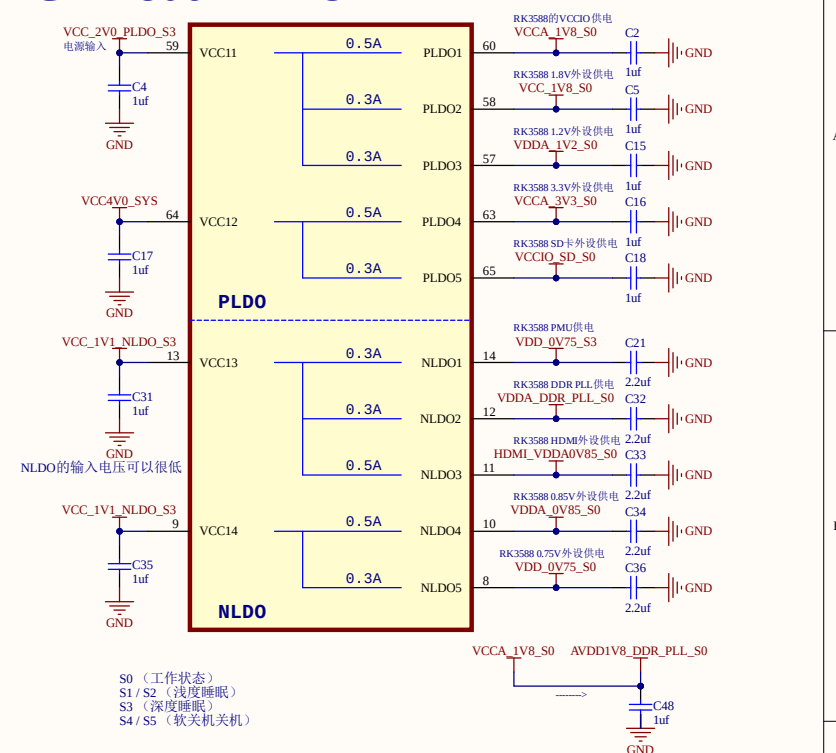


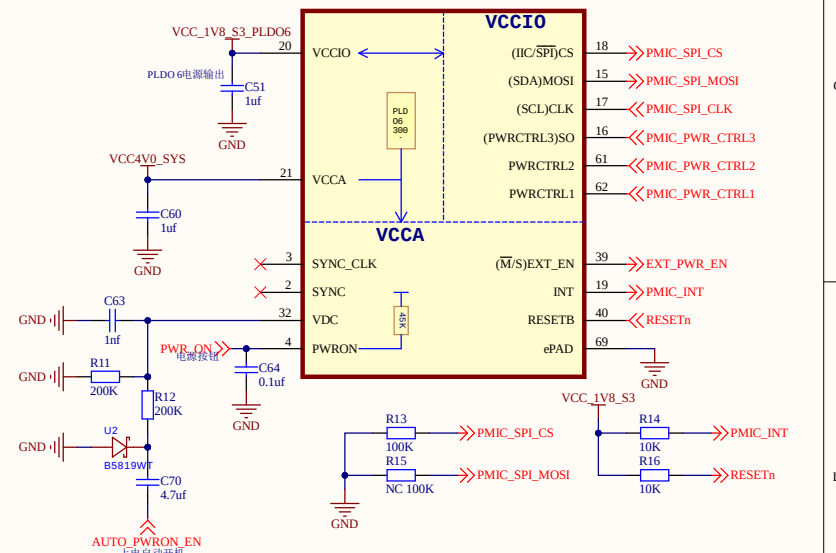
PMIC RK806-1 BUCK



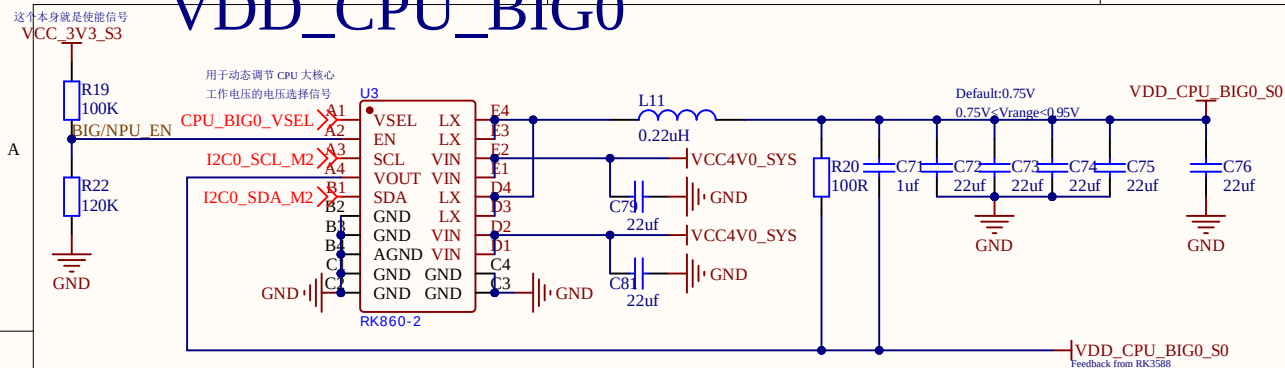
PMIC RK806-1 LDO



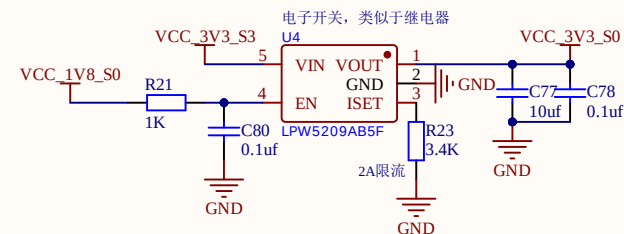
PMIC RK806-1 LDO



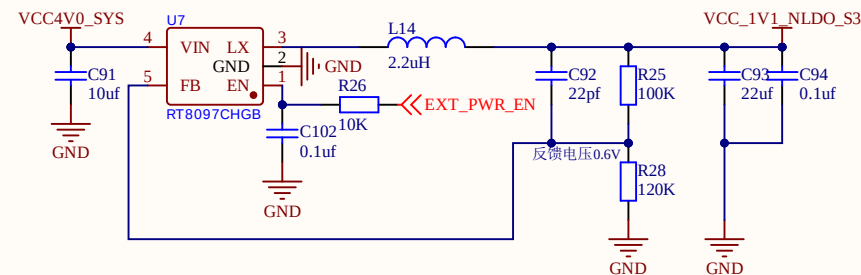
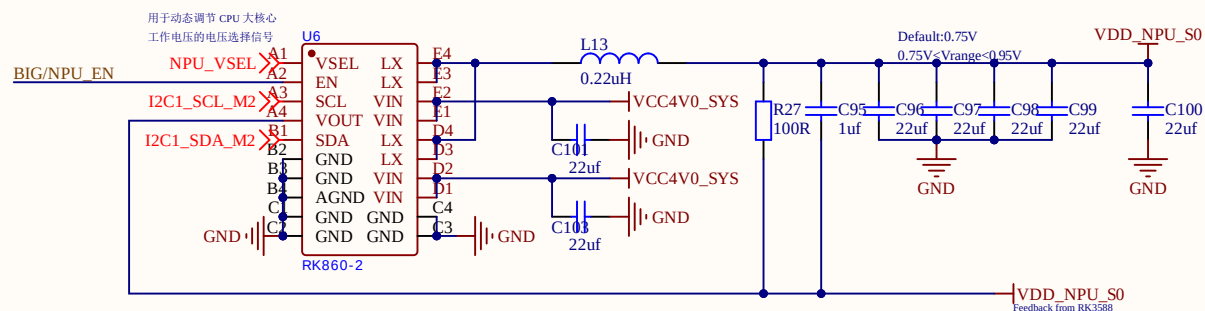
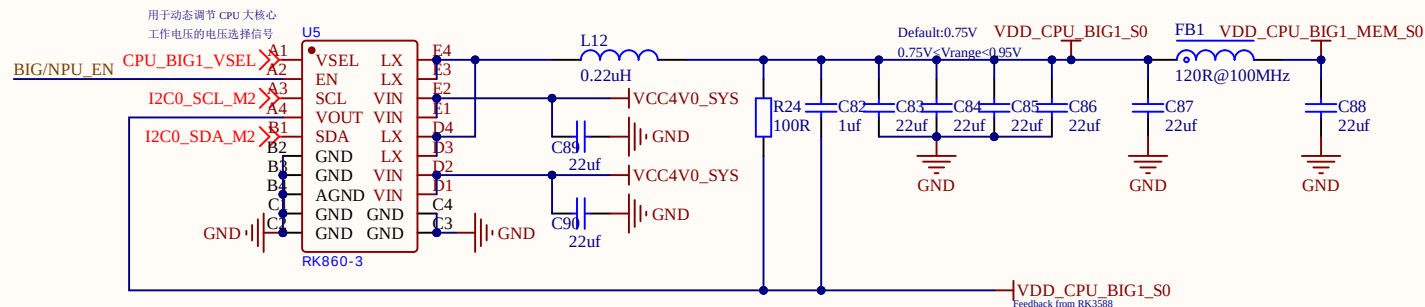
VDD_CPU_BIG0



VCC_3V3_S0



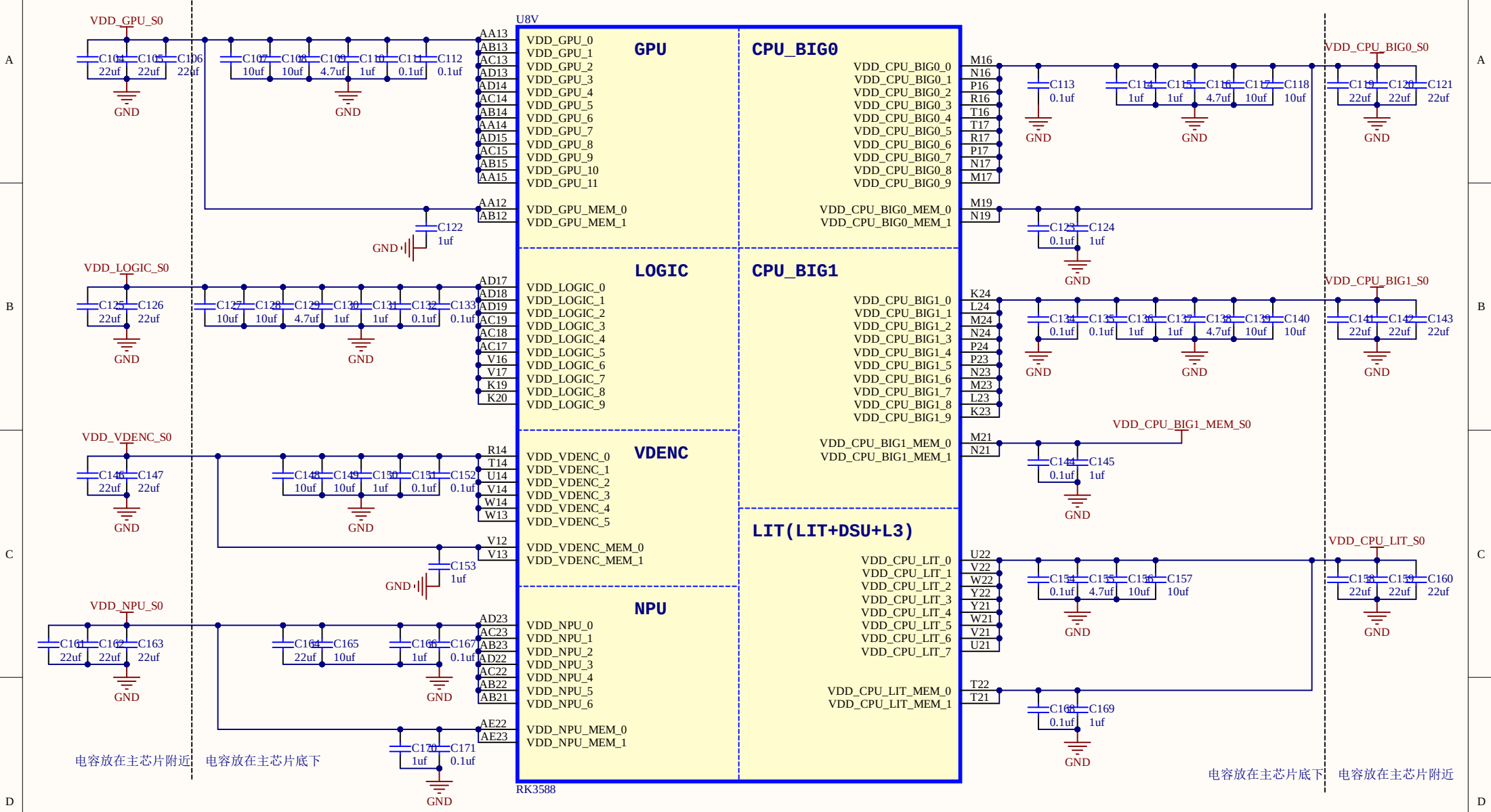
VDD_CPU_BIG1



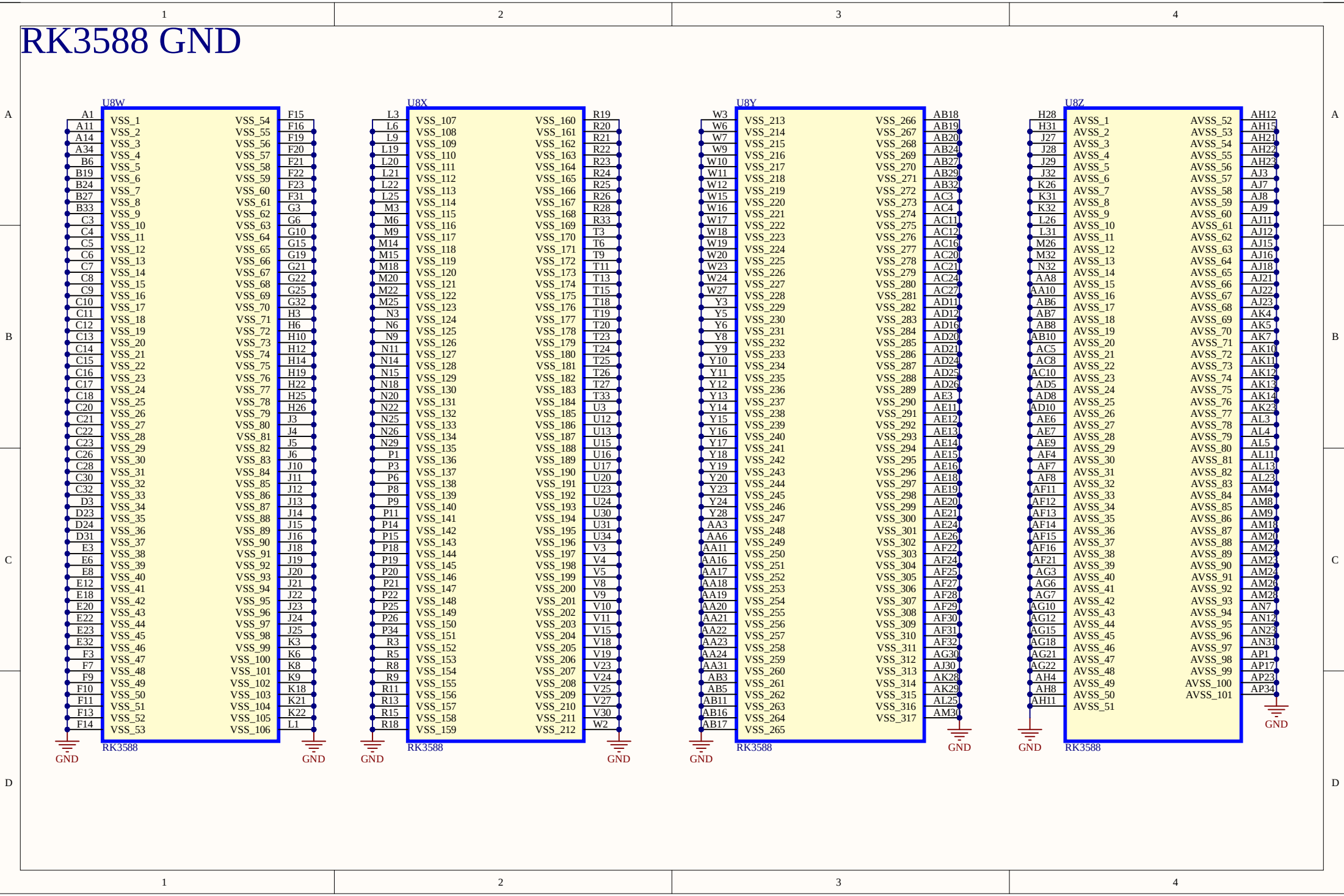
VCC_1V1_NLDO

VDD_NPU

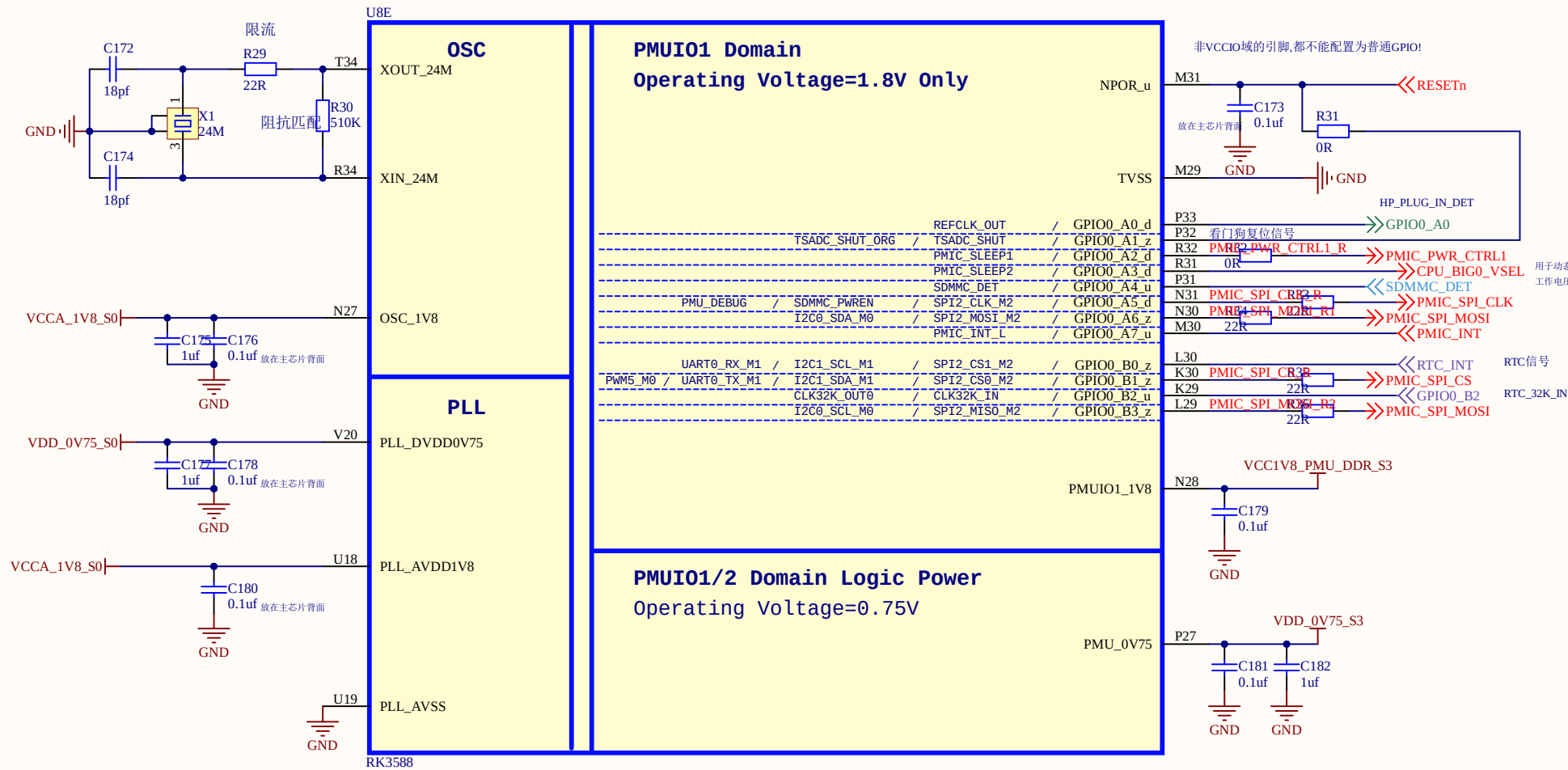
RK3588 POWER



RK3588 GND



RK3588 OSC/PLL/PMUIO1



RK3588 PMUIO2

电源管理、系统复位、基本初始化和安全启动相关

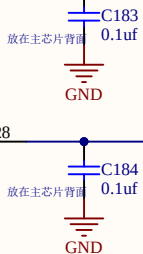
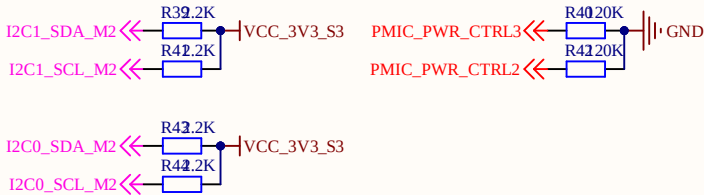
U8F

PMUIO2 Domain

Operating Voltage=1.8V/3.3V

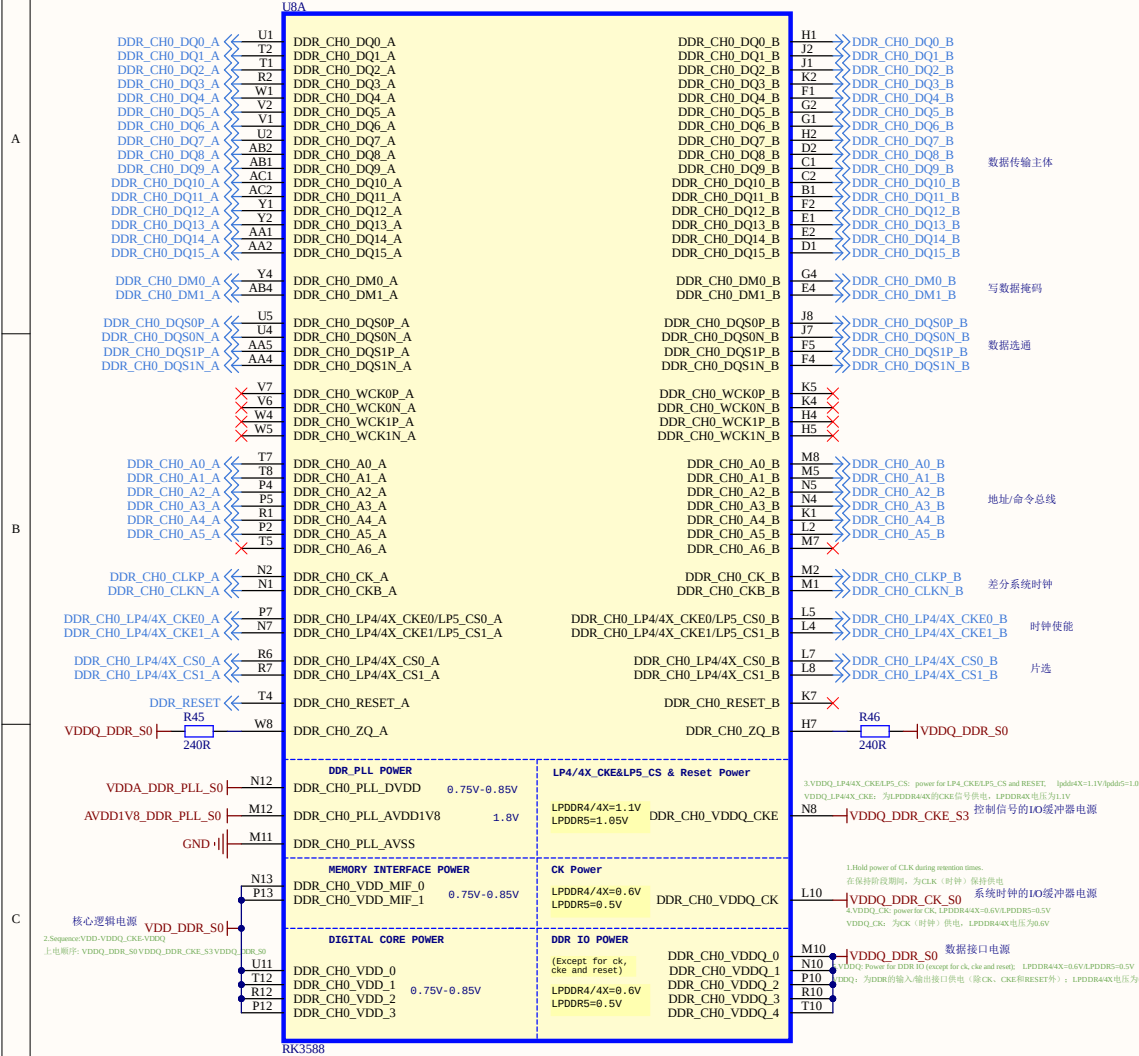
A		/	UART2_TX_M0	/		/	I2S1_MCLK_M1	/	PCIE30X1_1_CLKREQN_M0	/		/	I2C1_SCL_M0	/	JTAG_TCK_M2	/	GPIO0_B5_d	P29	需要串接100ohm电阻并加上tvs管	UART2_TX_M0_DEBUG	调试串口	A		
		/	UART2_RX_M0	/		/	I2S1_SCLK_M1	/	PCIE30X1_1_WAKEN_M0	/		/	I2C1_SDA_M0	/	JTAG_TMS_M2	/	GPIO0_B6_d	R29		UART2_RX_M0_DEBUG	调试串口			
		/		/	CAN0_TX_M0	/		/	I2S1_LRCK_M1	/	PCIE30X1_1_PERSTN_M0	/	SPI0_CS1_M0	/	I2C2_SCL_M0	/	PWM0_M0	/	GPIO0_B7_d	T28			GPIO0_B7	M2EKEY_PWR_EN
		/		/	CAN0_RX_M0	/	PDM0_CLK0_M1	/		/	PCIE30X1_0_CLKREQN_M0	/	SPI0_MOSI_M0	/	I2C2_SDA_M0	/	PWM1_M0	/	GPIO0_C0_d	T31			PCIE30X1_0_CLKREQN_M0	底板M2 E KEY控制信号
								/		/		/		/	PMIC_SLEEP3	/	GPIO0_C1_d	U32			PCIE30X1_0_CLKREQN_M0		底板M2 E KEY控制信号	
								/		/		/		/	PMIC_SLEEP4	/	GPIO0_C2_d	T32		PMIC_PWR_CTRL3_R	PMIC_PWR_CTRL3			
										/		/		/	PMIC_SLEEP5	/	GPIO0_C3_d	T30		PMIC_PWR_CTRL2_R	PMIC_PWR_CTRL2			
												/		/					R30		PCIE30X1_0_WAKEN_M0		底板M2 E KEY控制信号	
																			P30		PCIE30X1_0_PERSTN_M0		底板M2 E KEY控制信号	
																			T29		GPIO0_C6		M2EKEY_nWDISABLE	
B																			V31		GPIO0_C7	EARPHONE_CTL	B	
																			W31					
																			W30		I2C0_SCL_M2	底板RTC芯片IO扩展芯片和CAM0		
																			W29		I2C0_SDA_M2	底板RTC芯片IO扩展芯片和CAM0		
																			U33		GPIO0_D3	底板typec芯片FUSB302_INT		
																			V29		I2C1_SCL_M2	底板typec芯片		
																			V28		I2C1_SDA_M2	底板typec芯片		
																			W28		CPU_BIG1_VSEL			
C																								

RK3588



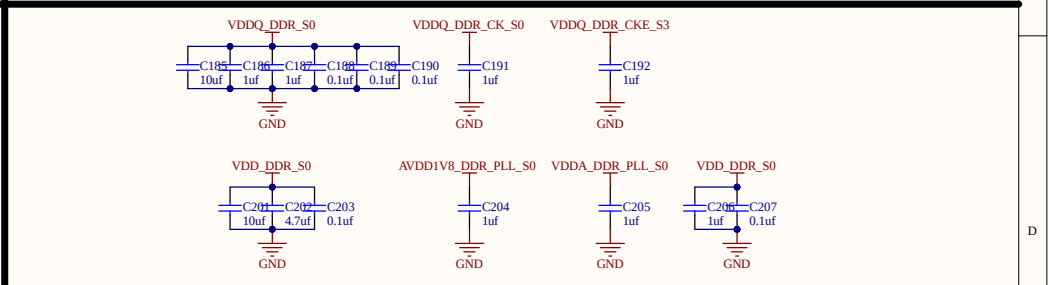
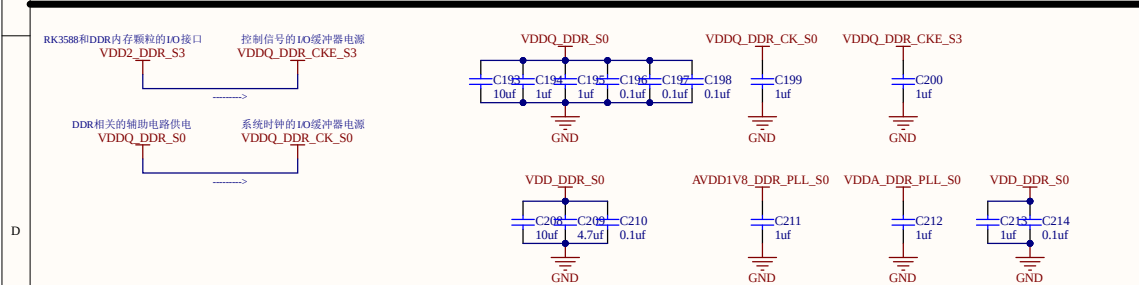
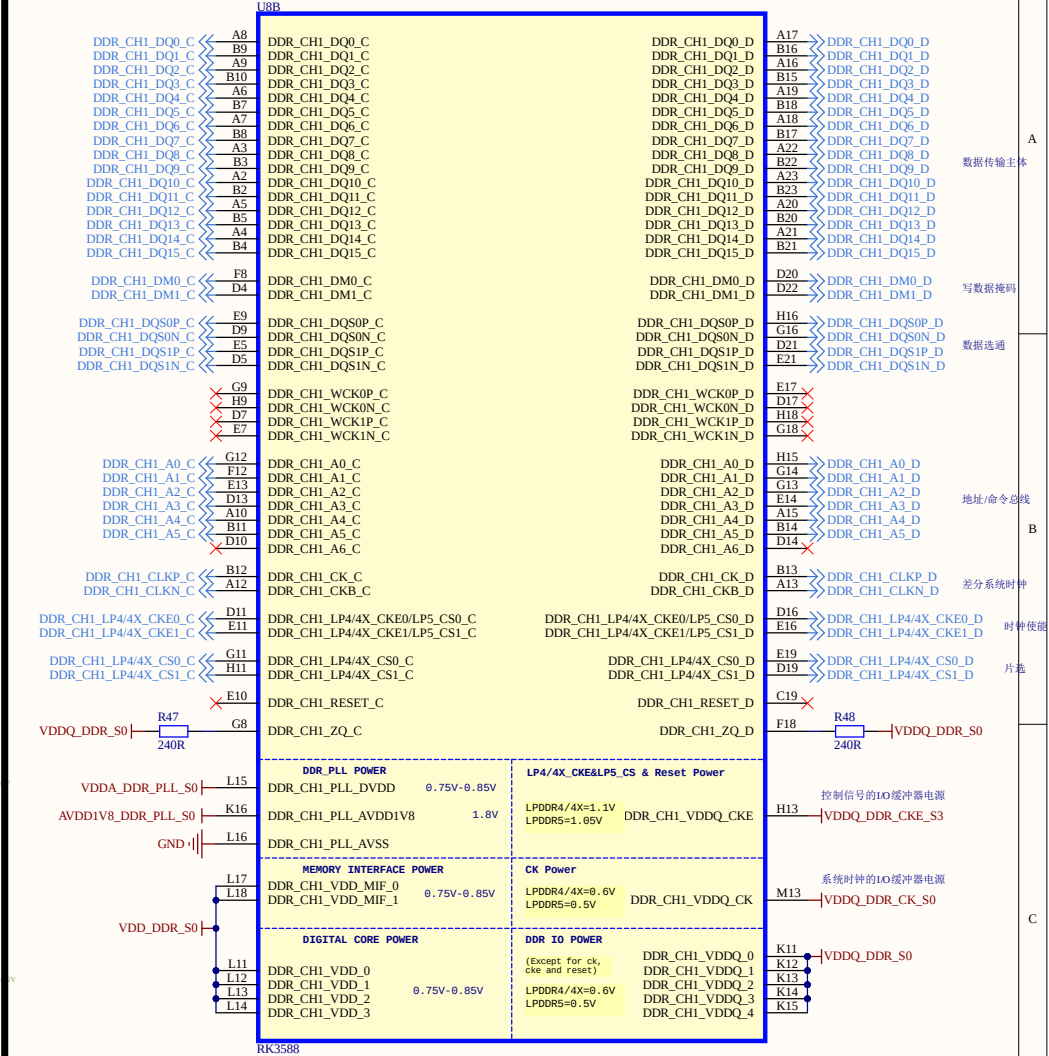
RK3588 DDR Controler CH0

DDR控制器接口



RK3588 DDR Controler CH1

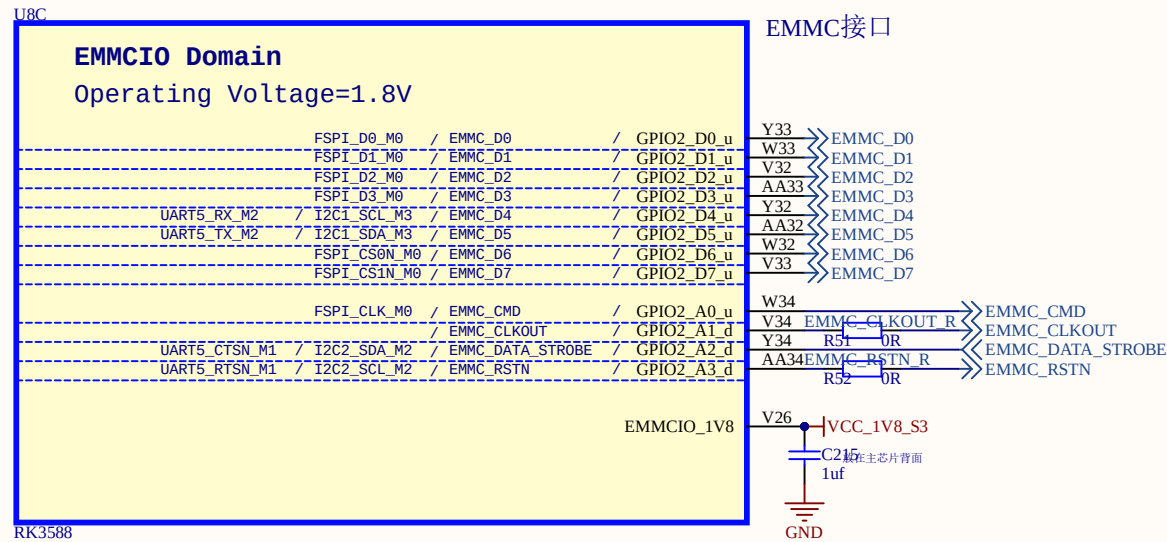
DDR控制器接口



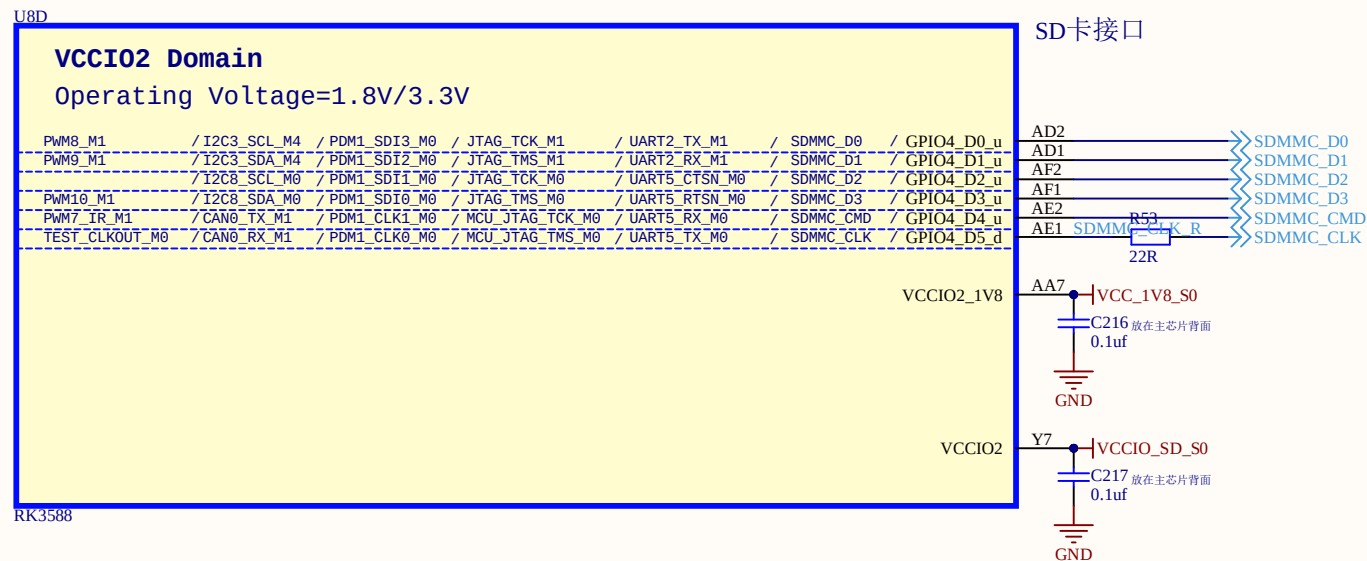
DDR Filter

DDR Filter

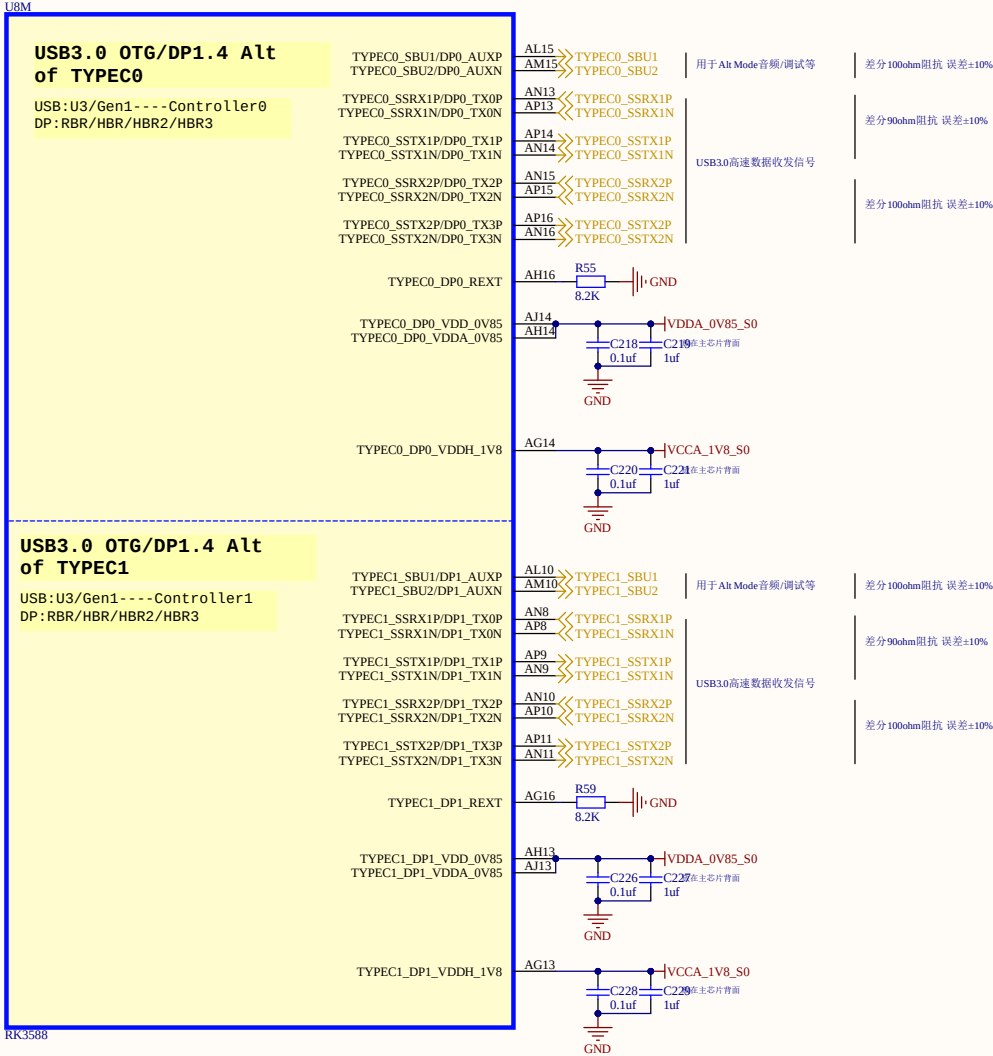
RK3588 EMMCIO Domain



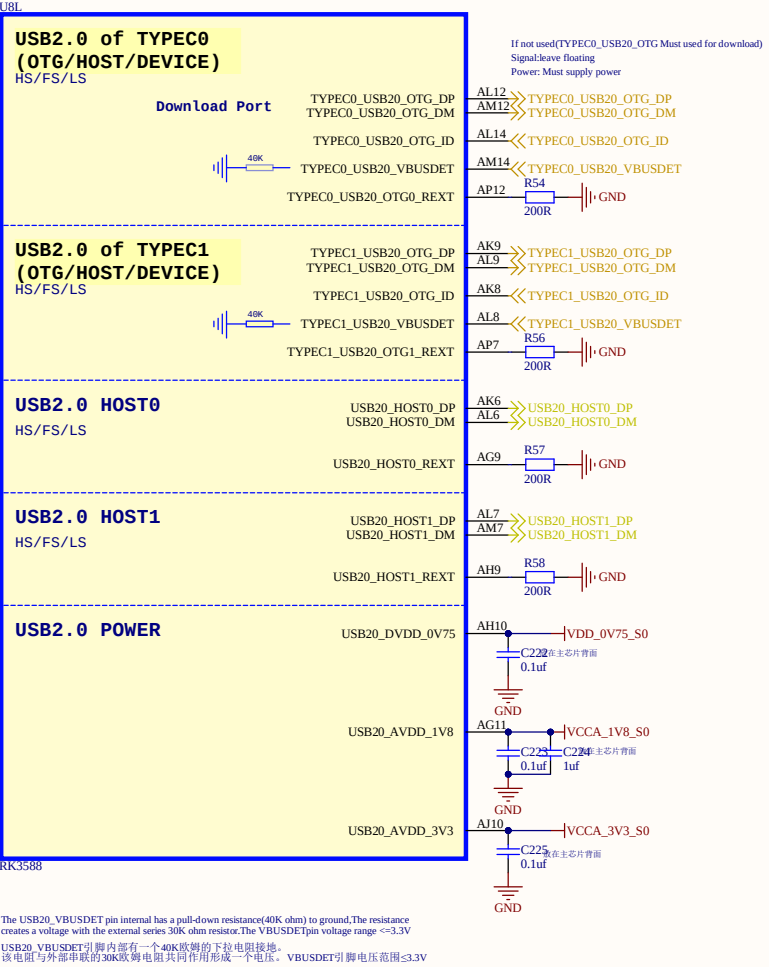
RK3588 VCCIO2 Domain



RK3588 TYPEC/DP

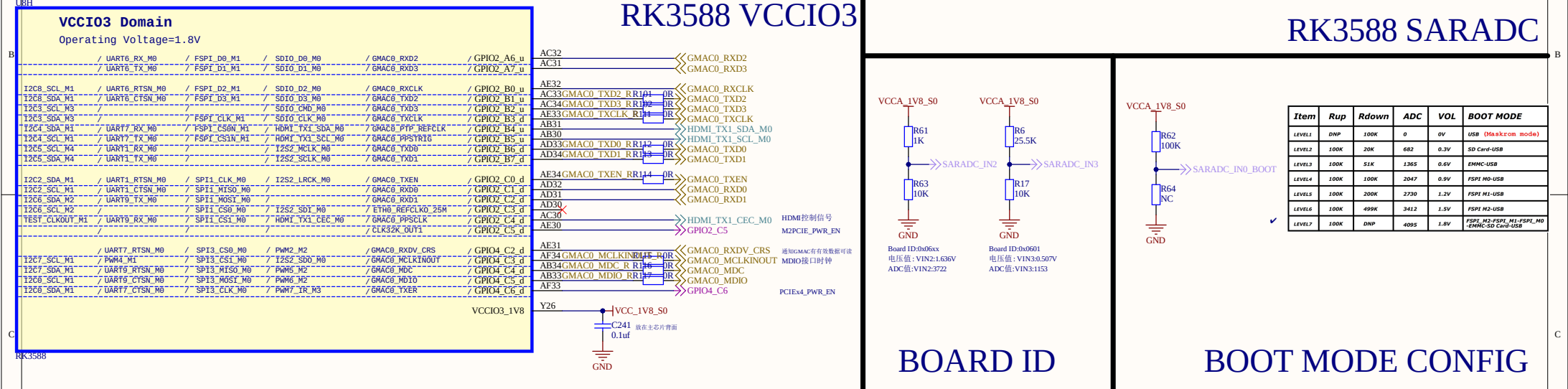
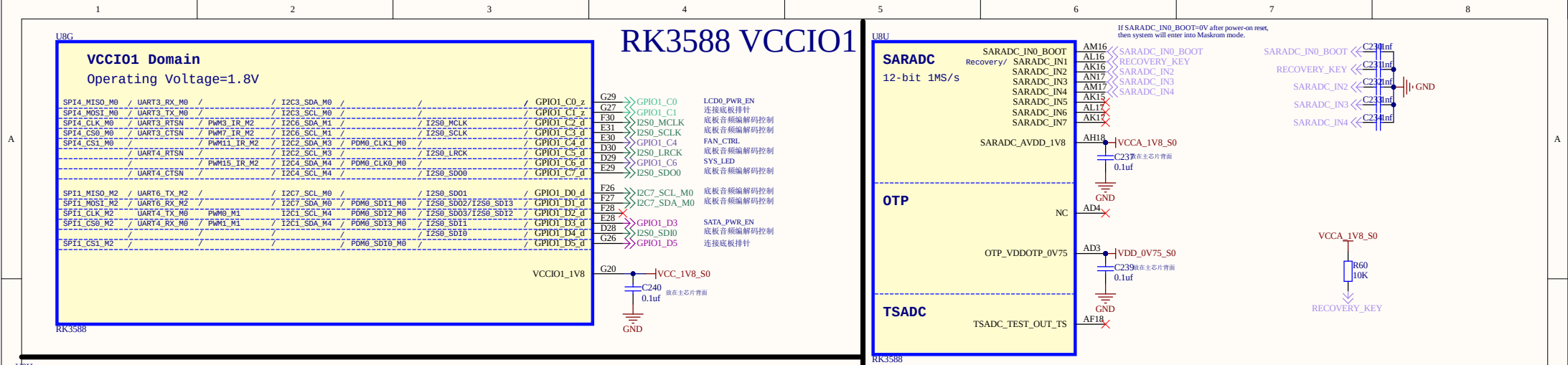


RK3588 USB2.0 HOST/OTG

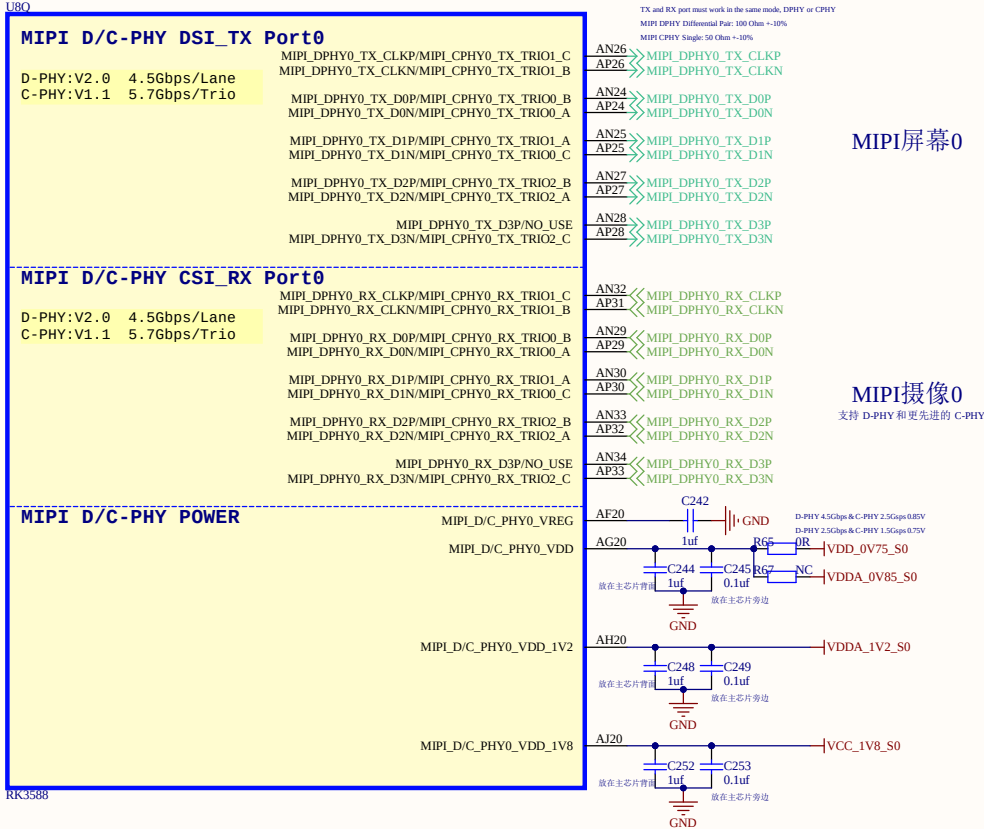


USB30/DP1.4 Alt Mode Configuration

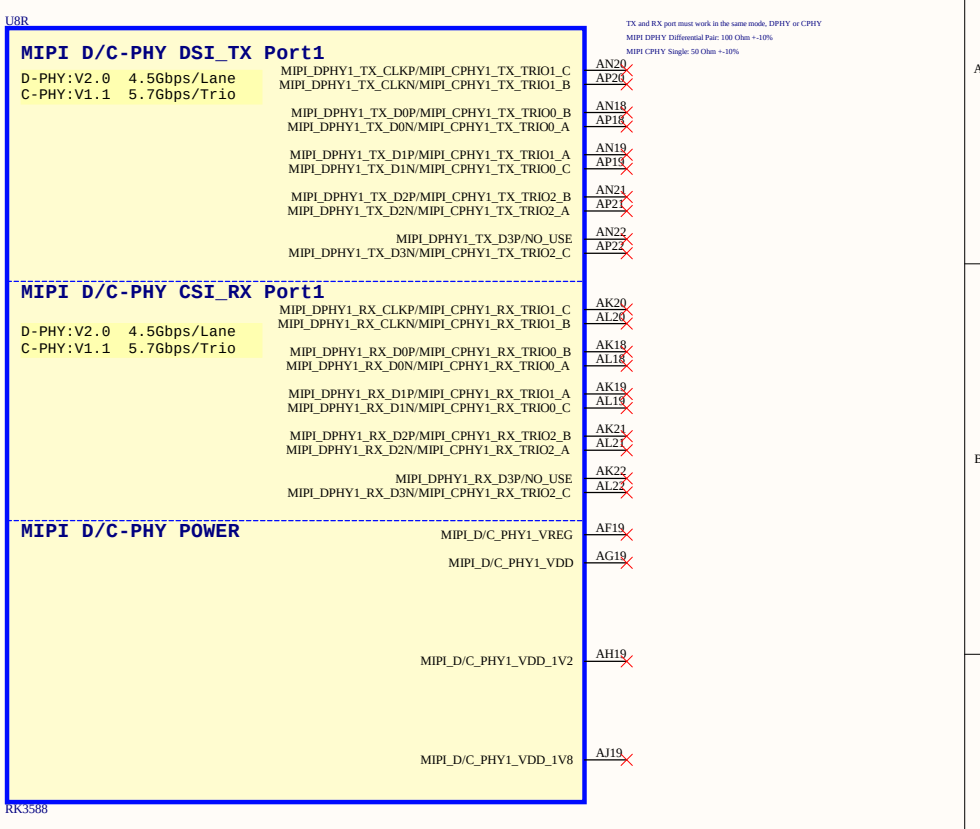
Option1	DP x4Lane	DP_TX_Lane0~3
Option2	USB30 x4Lane	DP_TX_Lane0~3
Option3	USB30X2Lane+DPX2Lane	USB30:Lane0 Lane1 DP:Lane2 Lane3
Option4	USB30X2Lane+DPX2Lane	USB30:Lane2 Lane3 DP:Lane0 Lane1



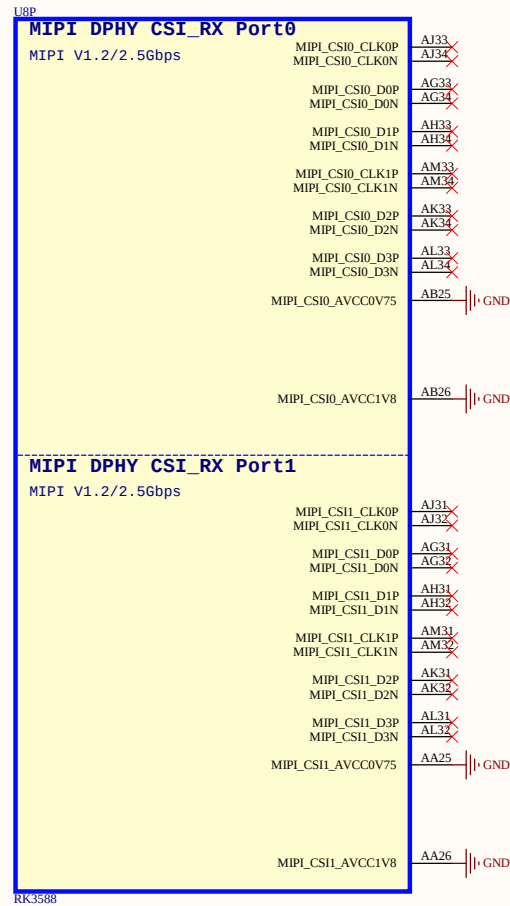
RK3588 MIPI_D/C_PHY0



RK3588 MIPI_D/C_PHY1



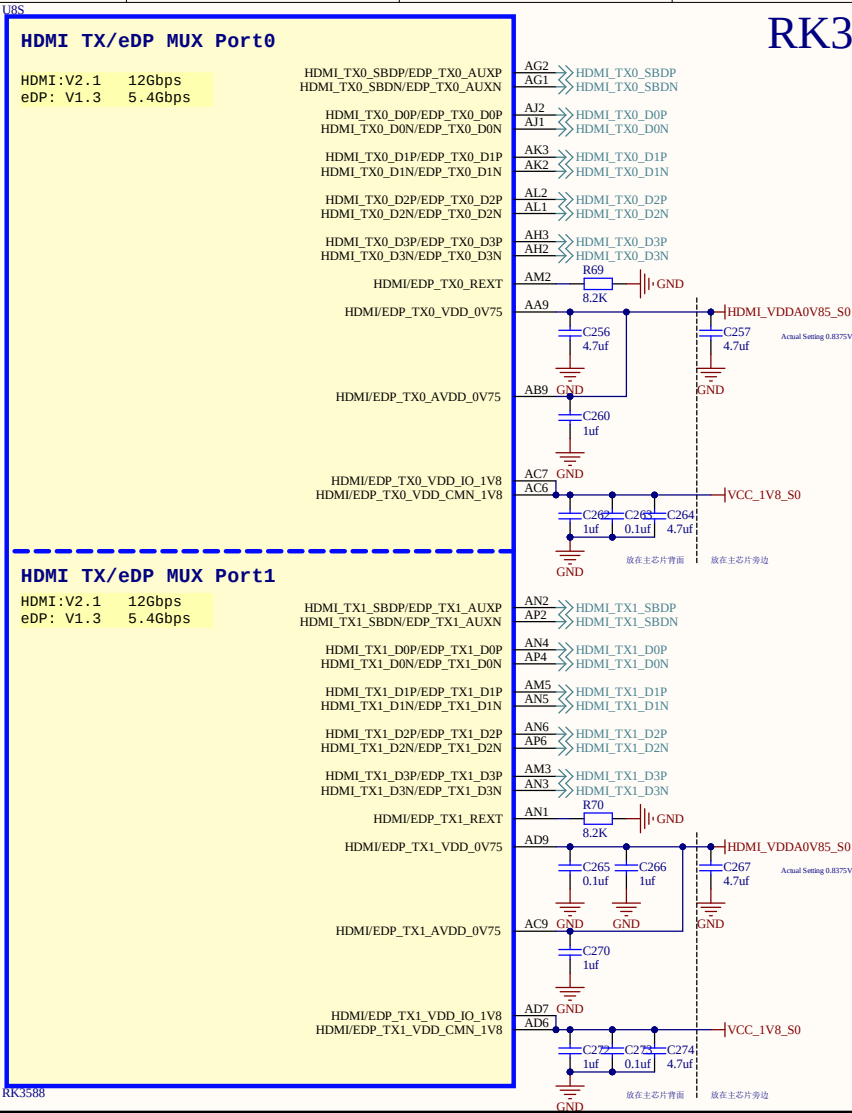
RK3588 MIPI_CSI_RX_PHY



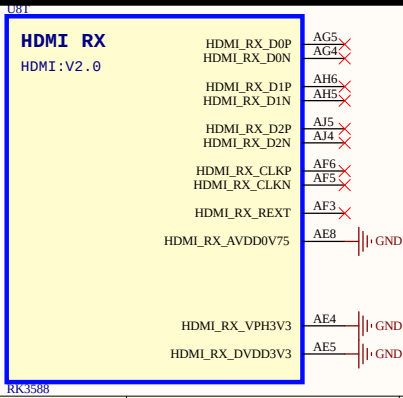
MIPI_CSI_RX Configuration

Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0
	Sensor2 x2Lane	MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

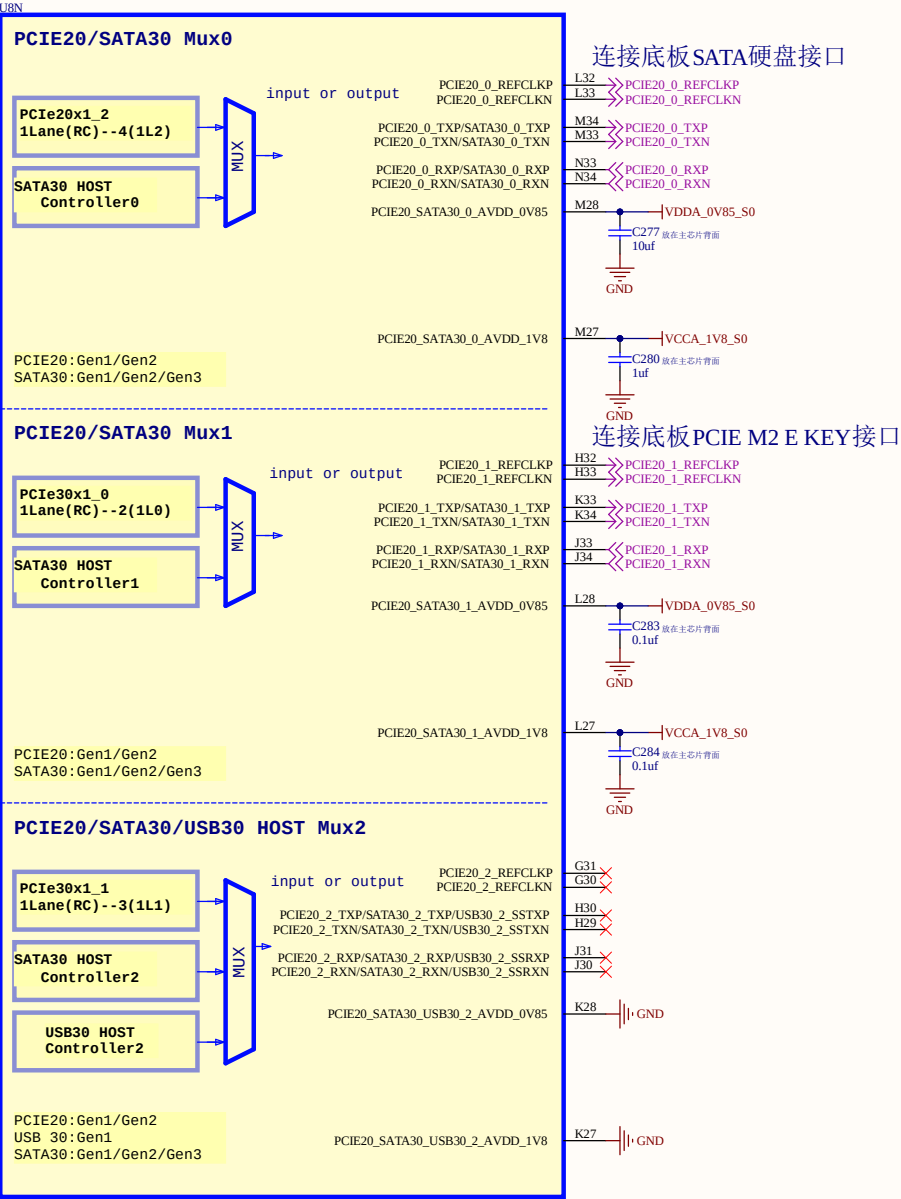
RK3588 HDMI2.1 TX



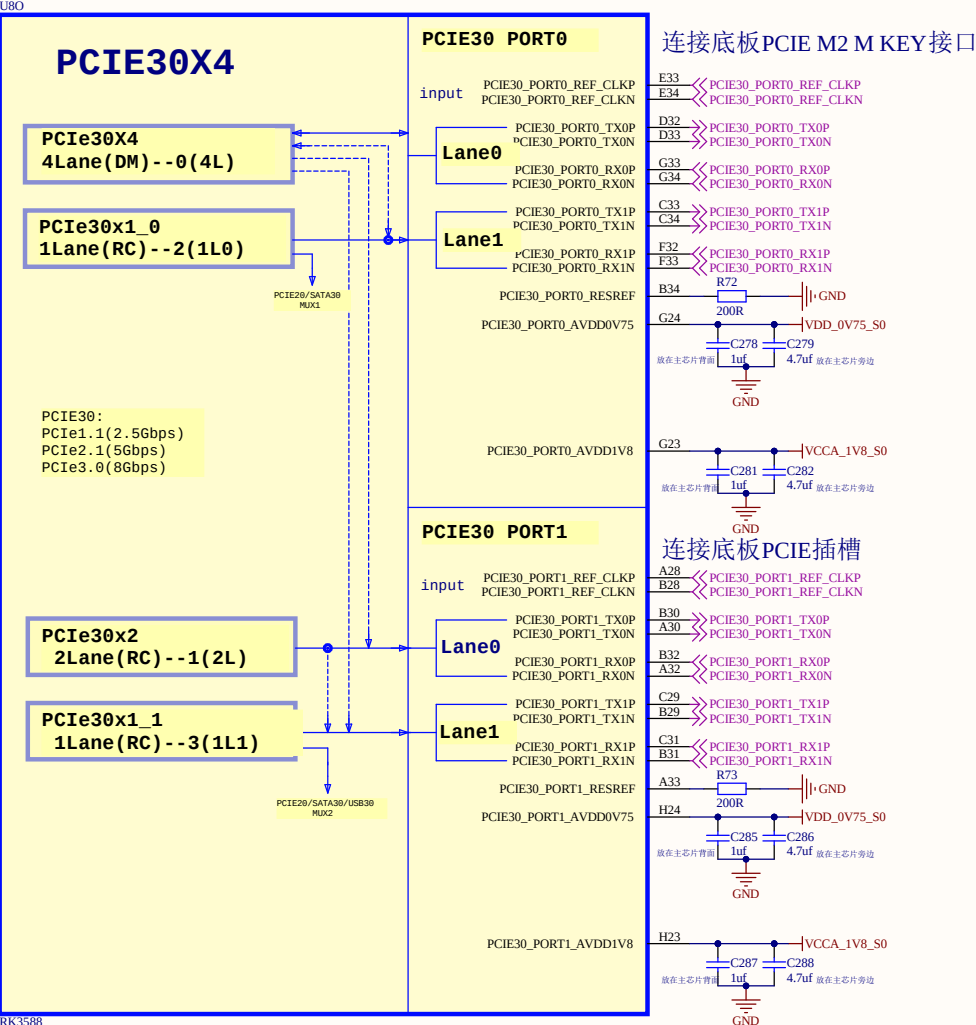
RK3588 HDMI2.0 RX



RK3588 PCIE2.0



RK3588 PCIE3.0



RK3588 VCCI04

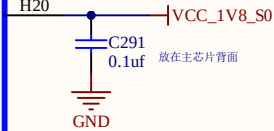
U8I

VCCI04 Domain

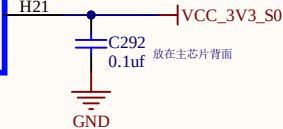
Operating Voltage=1.8V/3.3V

		/SPI4_MISO_M2	/PCIE30X1_1_CLKREQN_M2	/DP0_HPDIN_M2	/I2C2_SDA_M4	/UART6_RX_M1	/GPIO1_A0_d	A24	→I2C2_SDA_M4	MIPI DSI0 LCD接口
SATA1_ACT_LED_M1	/	/SPI4_MOSI_M2	/PCIE30X1_1_WAKEN_M2	/DP1_HPDIN_M2	/I2C2_SCL_M4	/UART6_TX_M1	/GPIO1_A1_d	A25	→I2C2_SCL_M4	MIPI DSI0 LCD接口
		/PWM0_M2	/SPI4_CLK_M2	/VOP_POST_EMPTY	/I2C4_SDA_M3	/UART6_RTSN_M1	/GPIO1_A2_d	A26	→GPIO1_A2	LCD0_BL_PWM
		/PWM1_M2	/SPI4_CS0_M2	/HDMI_TX1_SDA_M2	/I2C4_SCL_M3	/UART6_CTSN_M1	/GPIO1_A3_d	A27	✗	
		/	/SPI2_MISO_M0	/HDMI_TX1_SCL_M2	/	/	/GPIO1_A4_d	B25	→GPIO1_A4	HDMI0_TX_ON
		/	/SPI2_MOSI_M0	/HDMI_TX0_HPD_M0	/	/	/GPIO1_A5_d	B26	←HDMI_TX0_HPD_M0	
		/	/SPI2_CLK_M0	/HDMI_TX1_HPD_M0	/	/	/GPIO1_A6_d	C24	←HDMI_TX1_HPD_M0	
		/PWM3_IR_M3	/SPI2_CS0_M0	/PCIE30X1_1_PERSTN_M2	/PDM1_SDI0_M1	/	/GPIO1_A7_u	C25	→GPIO1_A7	HDMI1_TX_ON
		/	/SPI2_CS1_M0	/PCIE30X4_CLKREQN_M3	/PDM1_SDI1_M1	/	/GPIO1_B0_u	C27	→GPIO1_B0	TYPEC0_SBU1_DC
		/	/SPI0_MISO_M2	/PCIE30X4_WAKEN_M3	/PDM1_SDI2_M1	/	/GPIO1_B1_d	D25	→GPIO1_B1	TYPEC0_SBU2_DC
		/	/SPI0_MOSI_M2	/PCIE30X4_PERSTN_M3	/PDM1_SDI3_M1	/UART4_RX_M2	/GPIO1_B2_d	D26	←UART4_RX_M2	UART4_RX
SATA0_ACT_LED_M1	/	/SPI0_CLK_M2	/PCIE30X1_0_WAKEN_M2	/	/PDM1_CLK1_M1	/UART4_TX_M2	/GPIO1_B3_d	D27	→UART4_TX_M2	UART4_TX
		/	/SPI0_CS0_M2	/PCIE30X1_0_PERSTN_M2	/PDM1_CLK0_M1	/UART7_RX_M2	/GPIO1_B4_u	E24	←UART7_RX_M2	UART7_RX
		/	/SPI0_CS1_M2	/PCIE30X1_0_CLKREQN_M2	/	/UART7_TX_M2	/GPIO1_B5_u	E25	→UART7_TX_M2	UART7_TX
		HDMI_RX_HPDOUT_M2	/SPDIF0_TX_M0	/PCIE30X2_WAKEN_M3	/MIPI_CAMERA1_CLK_M0	/I2C5_SCL_M3	/UART1_TX_M1	E26	→I2C5_SCL_M3	连接底板排针
SATA2_ACT_LED_M1	/HDMI_RX_CEC_M2	/PWM13_M2	/SPDIF1_TX_M0	/PCIE30X2_PERSTN_M3	/MIPI_CAMERA2_CLK_M0	/I2C5_SDA_M3	/UART1_RX_M1	E27	→I2C5_SDA_M3	连接底板排针
		HDMI_RX_SCL_M2	/PWM14_M2	/	/MIPI_CAMERA3_CLK_M0	/I2C8_SCL_M2	/UART1_RTSN_M1	F24	→I2C8_SCL_M2	连接底板排针
		HDMI_RX_SDA_M2	/PWM15_IR_M3	/PCIE30X2_CLKREQN_M3	/MIPI_CAMERA4_CLK_M0	/I2C8_SDA_M2	/UART1_CTSN_M1	F25	→I2C8_SDA_M2	连接底板排针

VCCI04_1V8



VCCI04



RK3588

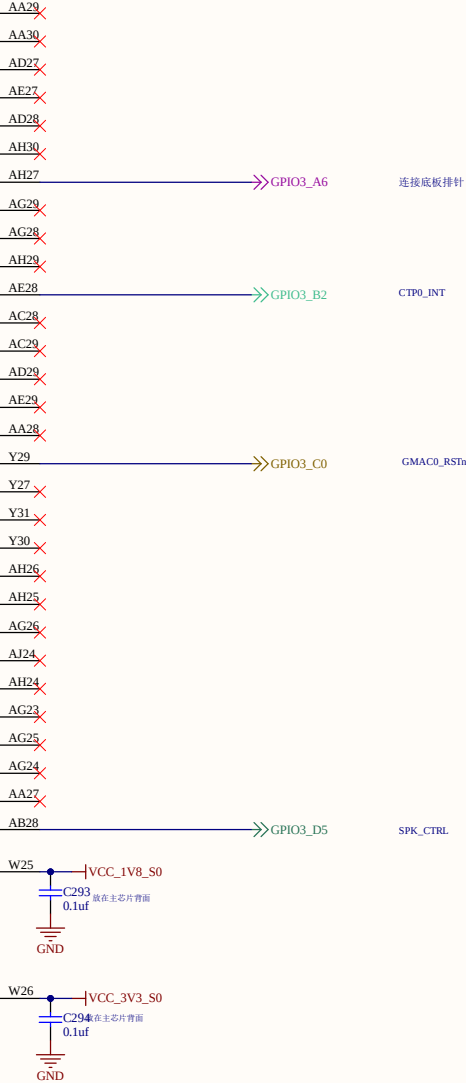
RK3588 VCCIO5

UBJ

VCCIO5 Domain									
Operating Voltage=1.8V/3.3V									
PWM0_M0	/ SPI4_MISO_M1	/	/ I2C6_SDA_M4	/ FSP1_D0_M2	/ I2S3_MCLK	/ SDIO_D0_M1	/ GMAC1_TXD2	/ GPIO3_A0_u	AA29
AUDDSM_LN	/ SPI4_MOSI_M1	/ PWM11_IR_M0	/ I2C6_SCL_M4	/ FSP1_D1_M2	/ I2S3_SCLK	/ SDIO_D1_M1	/ GMAC1_TXD3	/ GPIO3_A1_u	AA30
AUDDSM_LP	/ SPI4_CLK_M1	/ UART8_TX_M1	/	/ FSP1_D2_M2	/ I2S3_LRCK	/ SDIO_D2_M1	/ GMAC1_RXD2	/ GPIO3_A2_u	AD27
AUDDSM_RN	/ SPI4_CS0_M1	/ UART8_RX_M1	/	/ FSP1_D3_M2	/ I2S3_SDO0	/ SDIO_D3_M1	/ GMAC1_RXD3	/ GPIO3_A3_u	AE27
AUDDSM_RP	/ SPI4_CS1_M1	/ UART8_RTSN_M1	/	/	/ I2S3_SDI	/ SDIO_CMB_M1	/ GMAC1_TXCLK	/ GPIO3_A4_d	AD28
	/ MIPI_CAMERA0_CLK_M1	/ UART8_CTSN_M1	/ I2C4_SDA_M0	/ FSP1_CLK_M2	/	/ SDIO_CLK_M1	/ GMAC1_RXCLK	/ GPIO3_A5_d	AH30
	/ MIPI_CAMERA1_CLK_M1	/	/ I2C4_SCL_M0	/	/	/	/ ETH1_REFCLK0_25M	/ GPIO3_A6_d	AH27
PWM8_M0	/ MIPI_CAMERA2_CLK_M1	/	/	/	/	/	/ GMAC1_RXD0	/ GPIO3_A7_u	AG29
PWM9_M0	/ MIPI_CAMERA3_CLK_M1	/	/	/	/	/	/ GMAC1_RXD1	/ GPIO3_B0_u	AG28
PWM2_M1	/ MIPI_CAMERA4_CLK_M1	/ UART2_TX_M2	/	/	/	/	/ GMAC1_RXDV_CRS	/ GPIO3_B1_d	AH29
PWM3_IR_M1	/	/ UART2_RX_M2	/	/	/ I2S2_SDI_M1	/	/ GMAC1_TXER	/ GPIO3_B2_d	AE28
	/	/ UART2_RTSN	/	/	/ I2S2_SDO_M1	/	/ GMAC1_TXD0	/ GPIO3_B3_u	AC28
	/	/ UART2_CTSN	/	/	/ I2S2_MCLK_M1	/	/ GMAC1_TXD1	/ GPIO3_B4_u	AC29
PWM12_M0	/ CAN1_RX_M0	/ UART3_TX_M1	/	/	/ I2S2_SCLK_M1	/	/ GMAC1_TXEN	/ GPIO3_B5_u	AD29
PWM13_M0	/ CAN1_TX_M0	/ UART3_RX_M1	/	/	/ I2S2_LRCK_M1	/	/ GMAC1_MCLKINOUT	/ GPIO3_B6_d	AE29
	/	/	/ I2C3_SCL_M1	/ SPI1_MOSI_M1	/ HDMI_TX1_HPD_M1	/	/ GMAC1_PTP_REF_CLK	/ GPIO3_B7_d	AA28
	/	/ UART7_TX_M1	/ I2C3_SDA_M1	/ SPI1_MISO_M1	/	/	/ GMAC1_PPSTRIG	/ GPIO3_C0_d	Y29
	/ PCIE30X2_BUTTON_RSTN	/ UART7_RX_M1	/	/ SPI1_CLK_M1	/	/	/ GMAC1_PPSCCLK	/ GPIO3_C1_d	Y27
PWM14_M0	/	/ UART7_RTSN_M1	/ I2C8_SCL_M4	/ SPI1_CS0_M1	/	/ MIPI_TE0	/ GMAC1_MDC	/ GPIO3_C2_d	Y31
PWM15_IR_M0	/	/ UART7_CTSN_M1	/ I2C8_SDA_M4	/ SPI1_CS1_M1	/	/ MIPI_TE1	/ GMAC1_MDIO	/ GPIO3_C3_d	Y30
CAN2_RX_M0	/ PCIE30X4_CLKREQN_M2	/ UART5_TX_M1	/ FSP1_CS0N_M2	/ SPI3_CS0_M3	/ HDMI_TX1_CEC_M2	/	/ CIF_D8	/ GPIO3_C4_u	AH26
CAN2_TX_M0	/ PCIE30X4_WAKEN_M2	/ UART5_RX_M1	/ FSP1_CS1N_M2	/ SPI3_CS1_M3	/ HDMI_TX1_SDA_M1	/	/ CIF_D9	/ GPIO3_C5_u	AH25
	/ PCIE30X4_PERSTN_M2	/	/	/ SPI3_MISO_M3	/ HDMI_TX1_SCL_M1	/	/ CIF_D10	/ GPIO3_C6_u	AG26
	/ PCIE20X1_2_CLKREQN_M0	/	/ I2C5_SCL_M0	/ SPI3_MOSI_M3	/ HDMI_TX0_SCL_M2	/	/ CIF_D11	/ GPIO3_C7_u	AJ24
PWM8_M2	/ PCIE20X1_2_WAKEN_M0	/ UART4_RX_M1	/ I2C5_SDA_M0	/ SPI3_CLK_M3	/ HDMI_TX0_SDA_M2	/	/ CIF_D12	/ GPIO3_D0_u	AH24
PWM9_M2	/ PCIE20X1_2_PERSTN_M0	/ UART4_TX_M1	/	/ SPI0_MISO_M3	/ HDMI_RX_CEC_M1	/	/ CIF_D13	/ GPIO3_D1_d	AG23
	/ PCIE30X2_CLKREQN_M2	/ UART9_RTSN_M2	/ I2C7_SCL_M2	/ SPI0_MOSI_M3	/ HDMI_RX_SCL_M1	/	/ CIF_D14	/ GPIO3_D2_d	AG25
PWM10_M2	/ PCIE30X2_WAKEN_M2	/ UART9_CTSN_M2	/ I2C7_SDA_M2	/ SPI0_CLK_M3	/ HDMI_RX_SDA_M1	/	/ CIF_D15	/ GPIO3_D3_d	AG24
	/ PCIE30X2_PERSTN_M2	/ UART9_RX_M2	/	/ SPI0_CS0_M3	/ HDMI_RX_HPDOUT_M1	/ HDMI_TX0_HPD_M1	/ MCU_JTAG_TCK_M1	/ GPIO3_D4_d	AA27
PWM11_IR_M3	/ PCIE30X4_BUTTON_RSTN	/ UART9_TX_M2	/	/ SPI0_CS1_M3	/ DP1_HPDIN_M0	/	/ MCU_JTAG_TMS_M1	/ GPIO3_D5_d	AB28

VCCIO5_1V8

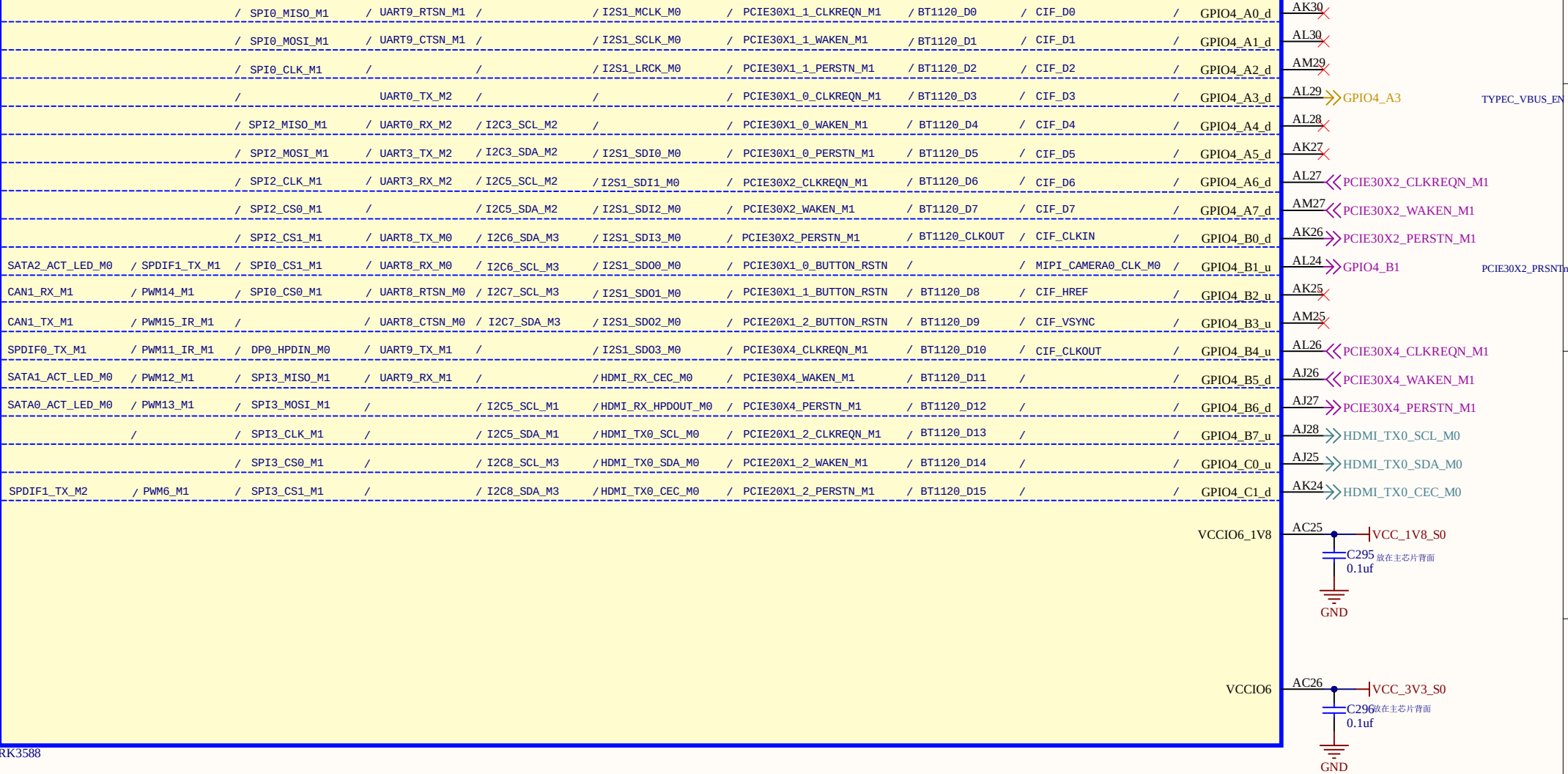
VCCIO5



RK3588 VCCIO6

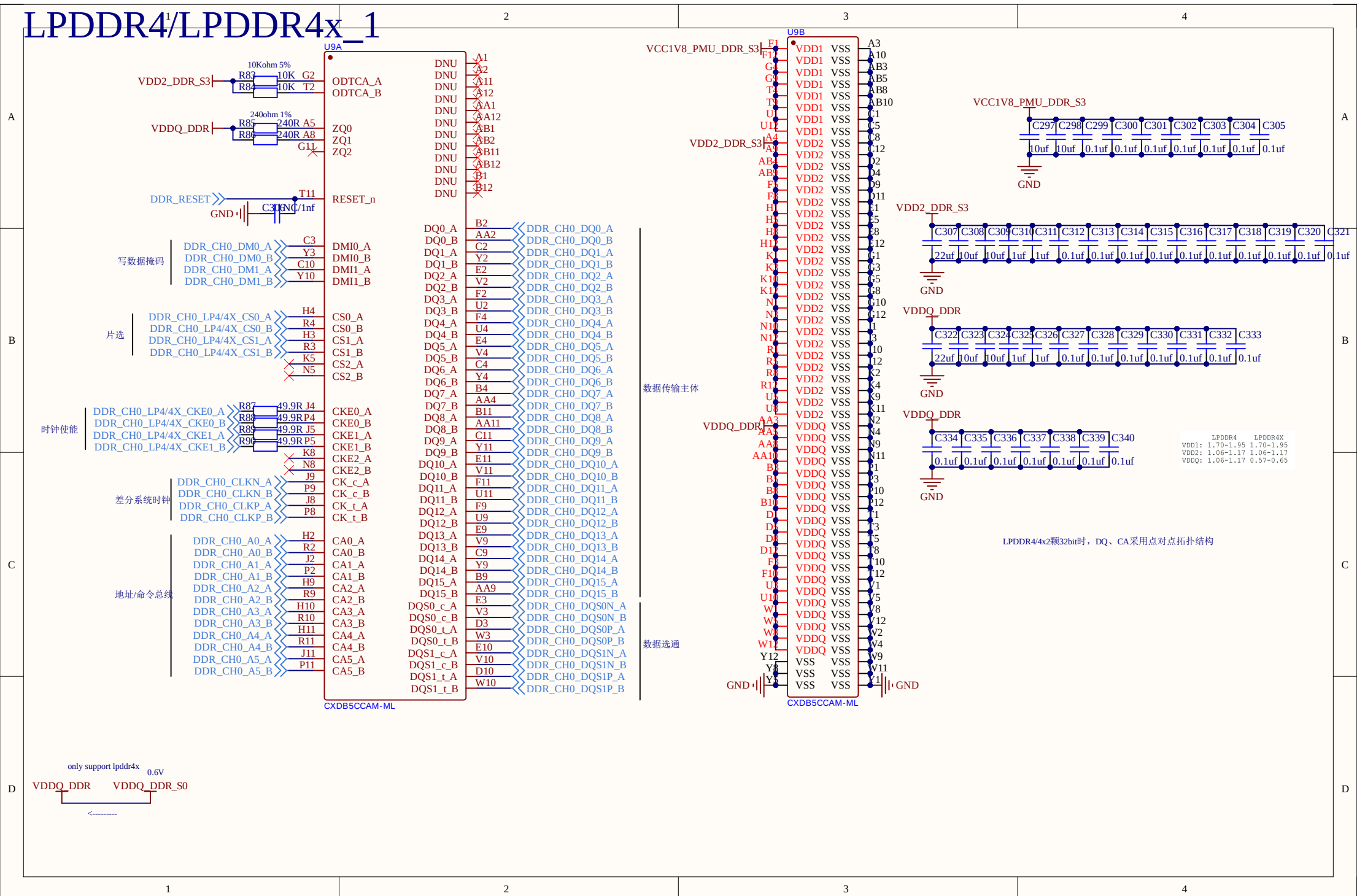
U8K

VCCIO6 Domain
Operating Voltage=1.8V/3.3V

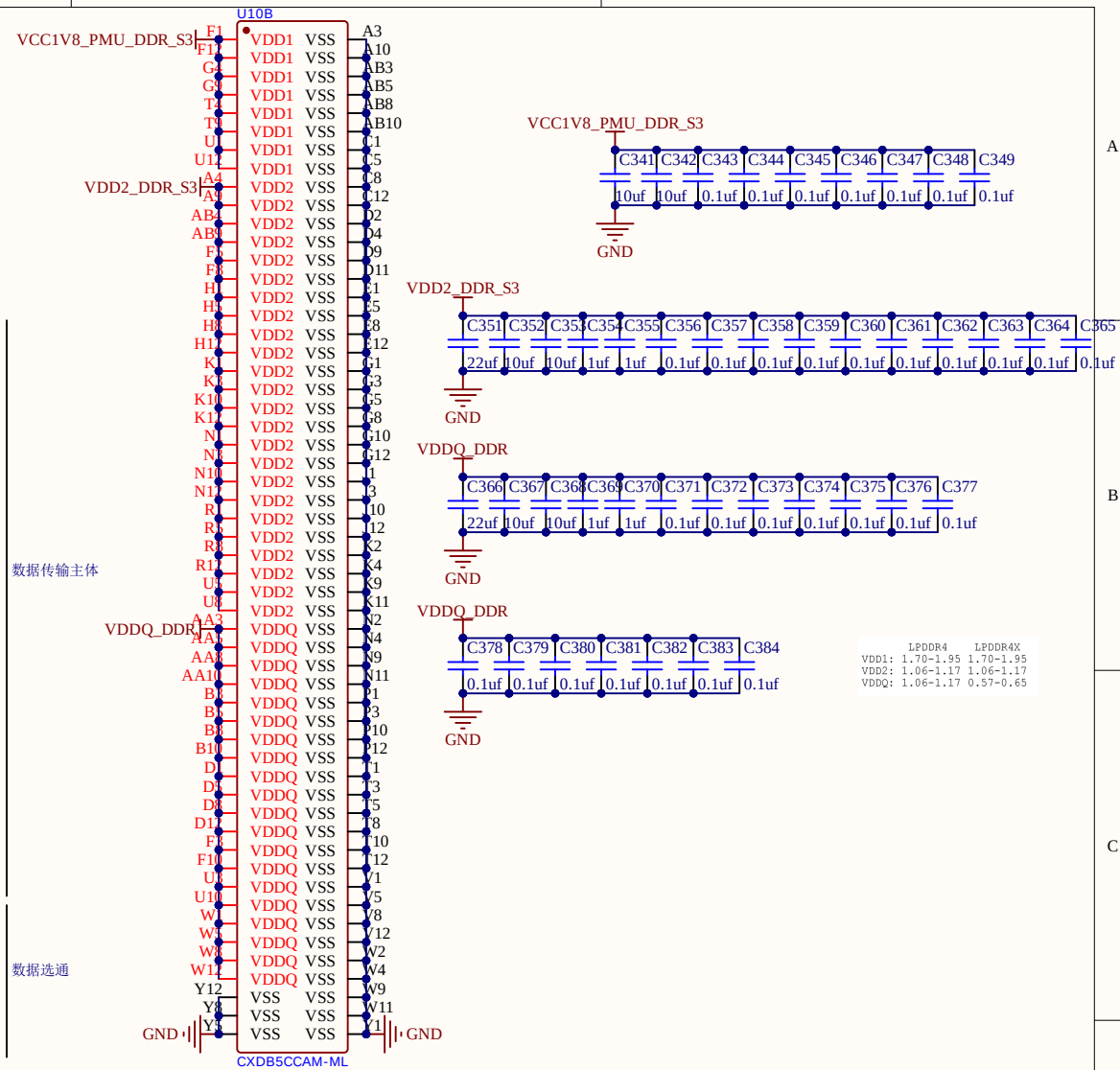
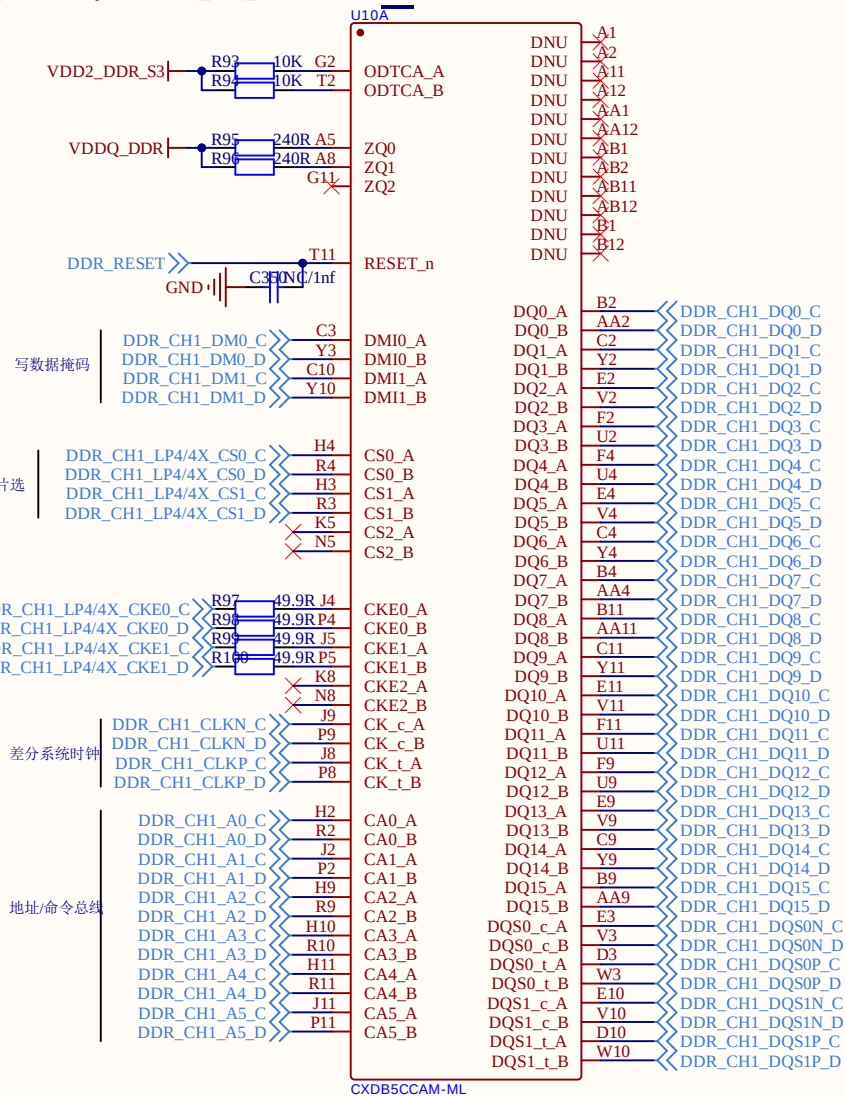


RK3588

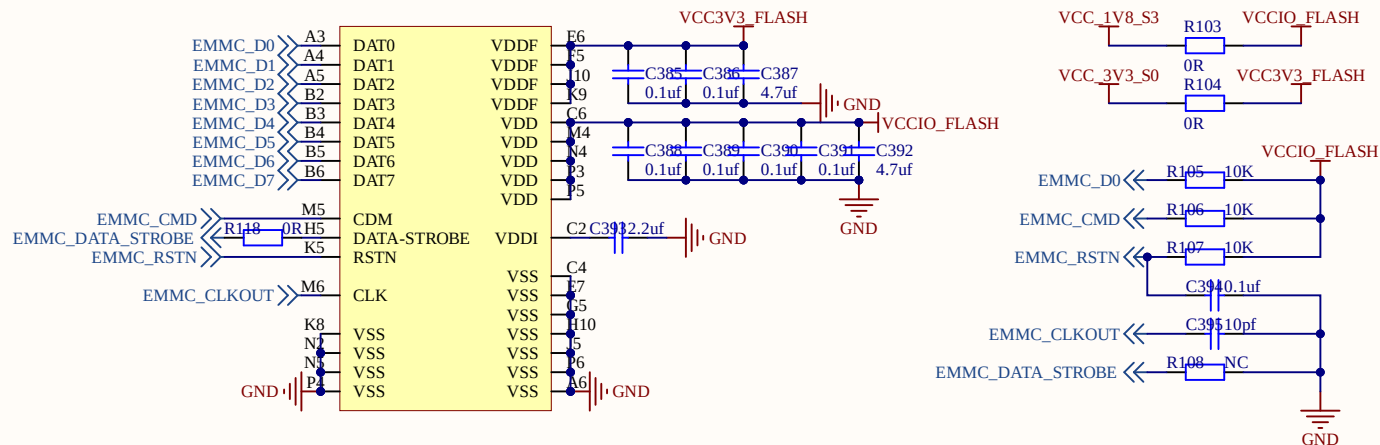
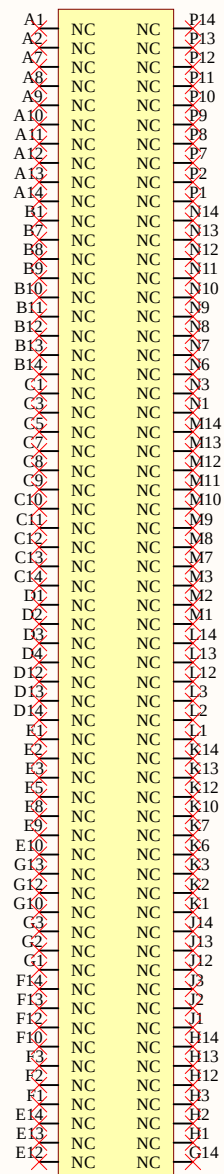
LPDDR4¹/LPDDR4x 1



LPDDR4¹/LPDDR4x 2



EMMC



SPI Flash

